

Quang Cuong DOAN

PhD, Department of Urban Planning and Design, The University of Hong Kong

✉ cuongdoan@connect.hku.hk ☎ +852 96029481

Research interests

Transport Geography, Spatial Analytics, GeoAI, Urban Vitality, Transit-Oriented Development.

Education

The University of Hong Kong (HKU), Hong Kong *Sept 2022 – Aug 2026*

Doctor of Philosophy in Urban Planning and Design

Field of study: Urban Analytics and Smart Cities

The University of Glasgow, United Kingdom *Oct 2025 – Nov 2025*

Visiting PhD at School of Geographical and Earth Sciences

Vietnam National University (VNU), Vietnam *2018 – 2020*

Master of Science in Land Administration (High Distinction; GPA: 3.89/4.0)

Vietnam National University (VNU), Vietnam *2013 – 2017*

Bachelor of Science in Land Administration (High Distinction; GPA: 3.68/4.0)

Professional Experience

University of Science, Vietnam National University (VNU-HUS) *Sept 2021 – Aug, 2022*

Academic advisor for Urban Development Management and Real Estate program

University of Science, Vietnam National University (VNU-HUS) *Sept 2018 – Aug, 2022*

Lecturer

Professional Affiliations

- Member, American Association of Geographers, USA
- Member, The International Association on Planning, Law, and Property Rights

Teaching Experience

PhD student *HKU*

As teaching assistant *Sept 2022 – May 2025*

- Research mentor
- Think global, Act local: You, Hong Kong and the World

Lecturer *VNU-HUS*

As Instructor *June 2018 – Aug 2022*

- Land registration
- Cadastre database management
- Land valuation

- Undergraduate Thesis
- Undergraduate's scientific research

Language

- Vietnamese (Mother tongue)
- English (Proficient)
- Chinese (Basic, level 2/6)

Technical skills

- Proficient in QGIS, ArcGIS
- SPSS, R and Python programming
- Object detection, Semantic segmentation

Research grants

1. Co-PI. "Revealing urban dynamics: Modelling interactions between land use, housing, transportation, and environmental systems using GeoAI". National Foundation for Science and Technology Development, Vietnam. 2025 – 2027.
2. PI. "Fuzzy-Multi-Criteria Analysis methods and GIS for providing land value information in Dong Anh District, Hanoi, VNU–University of Science, 2020 – 2021. 08/2020-08/2021.
3. Member. "Developing a set of criteria and indicators to integrate landscape ecology, regional linkages and climate change in land use planning to ensure sustainable development in Vietnam." Ministry of Science and Technology, 2021-2024.
4. Secretary. "An electronic land registration system in Tu Son town, Bac Ninh province, VNU –University of Science". 2019-2020.

Publications

Articles in Progress

1. Han, T.* , Wu, M., Chen, Y., Wang, M., Zhao, Q., **Doan, Q. C.** (2026). GeoAI for urban analytics: Opportunities and challenges. *Big Earth Data* (Accepted)

Journal articles

1. **Doan, Q. C.**, Bui, T. C. N.* (2026). Do better street configurations mean higher land prices? Nonlinear and spatial heterogeneity effects using GeoShapley. *Transportation Research Part A: Policy and Practice*. 212, 105134.
2. **Doan, Q. C.**, Wang, M., Zhang, X.* (2026). The impact of delayed urban rail projects on land prices in Hanoi, Vietnam. *Transport Policy*. 104277.
3. Tran, V. T., Nguyen, T. H. T., Dang, K. B., Do, T. T. T., Pham, T. P., Pham, S. L., Vu, K. H., Trinh, T. T. K., Do, T. T., **Doan, Q. C.** (2026). Developing a framework of land types for Vietnam's national land use planning: Integrating landscape ecology, regional linkage, and climate change. *Planning Practice & Research*, 1–25. <https://doi.org/10.1080/02697459.2026.2641972>
4. **Doan, Q. C.**, Zhang, X.* (2025). The costs of delay: Marginalization and sustainability challenges around Hanoi's urban rail projects. *Habitat International*, 166, 103594.
5. **Doan, Q. C.**, Zhang, X.* (2025). A systematic review of urban vitality studies: Trends and research opportunities. *Land Use Policy*, 159, 107745.

6. **Doan, Q. C.***, Ma, J., Chen, S., Zhang, X.* (2025). Nonlinear and threshold effects of the built environment, road vehicles, and air pollution on urban vitality. *Landscape and Urban Planning*, 253, 105204.
7. **Doan, Q. C.***, Vu, K. H., Trang, T. T. K., Bui, T. C. N. (2024). Examining the nonlinear threshold effects of the 5Ds built environment on land values using an interpretable machine learning model. *Journal of Geographical Sciences*, 34, 2509–2533.
8. **Doan, Q. C.*** (2024). The spatiotemporal trends of urban–rural green spaces and their heterogeneous relationships with population and economic vitality: Evidence from the Red River Delta, Vietnam. *Socio-Economic Planning Sciences*, 93, 101885.
9. **Doan, Q. C.**, Chen, C., He, S., Zhang, X.* (2024). How urban air quality affects land values: Exploring nonlinear relationships and threshold identification using explainable artificial intelligence. *Journal of Cleaner Production*, 434, 140340.
10. **Doan, Q. C.*** (2023). Determining the optimal land valuation model: A case study of Hanoi, Vietnam. *Land Use Policy*, 127, 106578.
11. **Doan, Q. C.*** (2023). Citizens' satisfaction with the land registration procedure: Evidence from Ha Dong District, Hanoi, Vietnam. *Survey Review*, 55, 471–480.
12. Tran, V. T., Pham, T. P., Nguyen, T. H. T., Do, T. T. T., Pham, S. L., Vu, K. H., **Doan, Q. C.**, Do, T. T. (2023). A set of criteria and indicators for integrating eco-landscape, regional linkages, and climate change into land use planning. *Journal of Cartography and Surveying Science*.
13. Nguyen, H. D., Vu, P. L., Ha, M. C., Dinh, T. B. H., Nguyen, T. H., Hoang, T. P., **Doan, Q. C.**, Pham, V. M., Dang, D. K. (2022). Flood susceptibility mapping using advanced hybrid machine learning and CyGNSS: A case study of Nghe An Province, Vietnam. *Acta Geophysica*, 70, 2785–2803.
14. Vu, P. L., Ha, M. C., Nguyen, P. B., Dinh, T. B. H., Pham, V. M., **Doan, Q. C.**, Nguyen, H. D. (2022). Application of GNSS reflectometry in water level monitoring using low-cost GNSS antennas: A case study in Tam Giang Lagoon, Thua Thien Hue Province. *Journal of Science: Earth and Environmental Sciences*. <https://doi.org/10.25073/2588-1094/vnuees.4878>
15. Nguyen, T. H. T., Thai, T. Q. N., Tran, V. T., Pham, T. P., **Doan, Q. C.**, Vu, K. H., Doan, H. G., & Bui, Q. T. (2020). Land consolidation at the household level in the Red River Delta, Vietnam. *Land*, 9(6), 196.

Conference papers

1. **Doan, Q. C.***, Thi Cam Ngoc Bui. (2026). Do better street configurations mean higher land prices? Nonlinear and spatial heterogeneity effects using GeoShapley. *GeoInformatic*, Singapore, July 18–22. (Oral presentation)
2. **Doan, Q. C.***, Zhang, X. (2026). The costs of delay: Marginalization and sustainable development challenges around Hanoi's urban rail projects. *Annual Meeting of the American Association of Geographers (AAG)*, San Francisco, USA, March 16–22. (Oral presentation)
3. **Doan, Q. C.***, Zhang, X. (2025). Land value uplift and decline: The complex causal effects of delayed urban rail projects. *Transportation Research Symposium*, Rotterdam, The Netherlands, May 25–28. (Poster presentation)
4. **Doan, Q. C.***, Zhang, X. (2025). Factors influencing citizens' intentions to cede land use rights for urban rail transit projects: An extended theory of planned behavior. *International Conference on Planning, Law and Property Rights*, Cardiff, United Kingdom, March 2–7. (Oral presentation)
5. **Doan, Q. C.***, Zhang, X. (2024). Examining the nonlinear relationships between streetscape characteristics, air quality, and pedestrian volume using machine learning techniques. *Annual Meeting of the American Association of Geographers (AAG)*, Honolulu, USA. (Oral presentation)

6. Tran, V. T.*, Pham, T. P., Nguyen, T. H. T., Do, T. T. T., Pham, S. L., Vu, K. H., **Doan, Q. C.**, Do, T. T. (2023). Integrating climate change factors into national land use planning and regional land resource allocation in Vietnam. *Proceedings of the 1st National Land Management Conference*, Vietnam National University of Agriculture, Hanoi, September.
7. **Doan, Q. C.*** (2021). Application of geographically weighted regression (GWR) and ordinary least squares (OLS) models in land valuation. *National Geography Conference*, Vietnam.
8. Vu, K. H.*, **Doan, Q. C.** (2021). Agricultural land conversion and its impacts on people's lives: A case study of Nha Be District, Ho Chi Minh City. *National Geography Conference*, Vietnam.
9. Do, T. T., Nguyen, C. H., Tran, V. T., Vu, K. H., & **Doan, Q. C.** (2019). Evaluating ecological adaptability and future trends of agricultural land use systems for spatial planning in Quoc Oai District, Hanoi. *FIG Working Week*, Hanoi, Vietnam.
10. **Doan, Q. C.***, Tran, V. T., Pham, L. T. (2018). Application of GIS and multi-criteria analysis to assess spatial suitability of industrial land in Dong Anh, Hanoi. *National Geography Conference*, Danang, Vietnam.
11. **Doan, Q. C.***, Tran, V. T., Pham, L. T. (2017). Constructing a land use planning database: A case study of Dong Anh District, Hanoi. *Vietnam National GIS Conference*, Quy Nhon, Vietnam.

Book chapter

1. **Quang Cuong Doan***. (2021). GIS and Fuzzy Logic Approach for Providing Land Value Information: A Case Study in Hanoi City. In: *Huong, L.T.T., Pomeroy, G.M. (eds) AUC 2019. Advances in 21st Century Human Settlements*. Springer, Singapore.

Invited talk

1. Toward sustainable urban design: Effects of traffic vehicles and urban design characteristics on pedestrian volumes. *University of Science, Vietnam National University*. Hanoi, Vietnam, October 2024.

Honors and Awards

- Additional funding from Research Grant Council, Hong Kong, 2026
- Finalist of The Tony Crook Award for Early Career Research category of the RTPi Awards for Research Excellence 2025
- Additional funding from Research Grant Council, Hong Kong, 2025
- HKU Foundation First Year Excellent PhD Award 2024/25
- Glasgow/HKU Early Career Mobility Funds, 2025 - 2026
- Professor and Mrs Ko Wah Man Norman Enlightenment Scholarship, 2025
- HKU Foundation Publication Award for Research Postgraduate Students, 2024
- Additional funding from Research Grant Council, Hong Kong, 2023
- Hong Kong PhD Fellowship Award (Research Grants Council, Hong Kong), 2022 - 2026
- HKU - Presidential PhD Scholarship Award, 2022 - 2026
- Scholarship for excellent master student (ManManMan organization, Korea), 2019
- Scholarship for excellent student (Bank for Investment and Development of Vietnam), 2019
- Third prize, Students' scientific research competition (as supervisor), 2019
- Vietnam National University Outstanding Graduate Award, 2017
- Second Prize, Students' scientific research competition, Vietnam National University, 2017
- Scholarship for excellent student (Bank for Investment and Development of Vietnam), 2016

- Scholarships for Excellent Academic Performance Student, 2013-2017
- Consolation prizes in Geography competition for excellent students in 2012.

Selected services

Reviewer for academic journals

- Computers, Environment and Urban Systems
- Land Use Policy
- Sustainable Cities and Society
- Applied Geography
- International Journal of Digital Earth
- Journal of Housing and the Built Environment
- Journal of Planning Literature
- Environment and Planning B: Urban Analytics and City Science
- Applied Spatial Analysis and Policy
- City and Environment Interactions
- Frontiers of Architectural Research
- Human Ecology
- Geoinformatics for Spatial-Infrastructure Development in Earth and Allied Sciences - Proceedings of GIS-IDEAS 2023

References

Prof. Xiaohu Zhang

- Assistant Professor, Department of Urban Planning and Design, The University of Hong Kong
- Email: zhangxh@hku.hk

Prof. Shenjing He

- Professor, Head of Department of Urban Planning and Design, The University of Hong Kong.
- Email: sjhe@hku.hk

Prof. Wang Mingshu

- Reader, School of Geographical and Earth Sciences, The University of Glasgow, UK
- Email: Mingshu.Wang@glasgow.ac.uk

Prof. Jun Ma

- Associate Professor, Department of Urban Planning and Design, The University of Hong Kong
- Email: junma@hku.hk